DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE – RAIGAD -402 103

Supplementary Examination – Dec - 2022

Branch: Electronics and Telecommunication Engineering **Sem.: - IV**

Subject with Subject Code: - Signals and Systems (EC12) **Marks: 60**

Date:- **Time:- 3 Hr.**

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions from Q. 1 to Q. 6.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

(Marks)

Q.1 A) Calculate and draw even and odd parts of the signal: (04)
$x(t) = e^{-2t}\cos(t)$

B) Compute the energy and power of the signal $x(n) = a^n u(n)$ where $|a| < 1$. (04)

C) Check linearity for the system given by equation: $y(n) = 2x(n)-3$ (04)

Q.2 A) Find the convolution of a rectangle signal or gate function $x(t)$ as shown in fig. 1 with itself. (08)

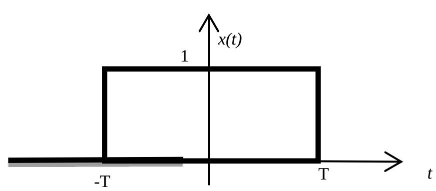

Fig. 1 : Given function $x(t)$

B) Function $x(t)$ is as shown in fig. 2. Find a) $x(2t)$ b) $x(-t+2)$ (04)

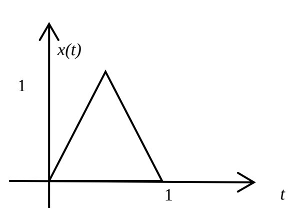

Fig. 2 : Given function $x(t)$

Q. 3 A) State and explain the Dirichlet conditions for the existance of Fourier series. (06)

B) Find exponential Fourier series for the waveform $x(t)$ shown in the fig. 3. (06)

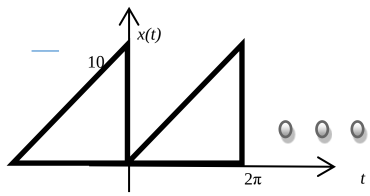

Fig. 3 : Given function $x(t)$

Q. 4 A) Find and sketch the Fourier transform of the impulse train $x(t) = \delta_{To}(t)$ depicted in figure 4. (06)

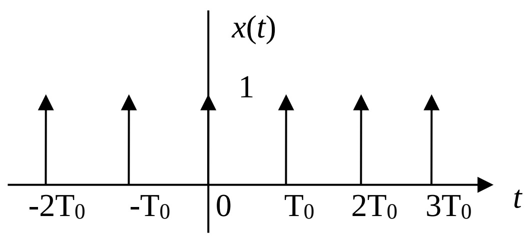

Fig. 4 : Given function $x(t)$

B) Determine the Laplace transform of $x(t) = e^{-2t}u(t) - e^{-3t}u(t)$. (06)
Also depict the ROC and the locations of poles and zeros in the s-plane.

Q.5 A) For the following system functions, find $h(n)$ and check whether the corresponding system is causal, anticausal, or noncausal. (09)

1. $H(s) = \frac{1}{s^2+5s+6}$ R{s} > -2
2. $H(s) = \frac{1}{s^2+5s+6}$ R{s} < -3
3. $H(s) = \frac{1}{s^2+5s+6}$ -2 < R{s} < -3

B) Determine the z-transform of the sequence: (03)

$$g(n) = na^n u(n)$$

using appropriate property of the z-transform.

Q.6 A) Determine the Nyquist rate for the following signals: (06)

i) $x(t) = \sin(200\pi t)$

ii) $x(t) = \sin2(200\pi t)$

iii) $x(t) = 1 + \cos(200\pi t) + \sin(400\pi t)$

iv) $x(t) = \cos(150\pi t)\sin(100\pi t)$

B) State and explain Bayes' Theorem. Solve the following example with the help of Bayes' theorem. (06)

In a factory, four machines A_1, A_2, A_3 and A_4 produce 10%, 20%, 30% and 40% of the items, respectively. The percentage of defective items produced by them is 5%, 4%, 3% and 2%, respectively. An item selected at random is found to be defective. What is the probability that it was produced by the machine A_2?